

# ZeroPad3d#

<https://pytorch.org/docs/stable/generated/torch.nn.ZeroPad3d.html>

## Description:

Pads the input tensor boundaries with zero. For N-dimensional padding, use `torch.nn.functional.pad()`. `padding` (int, tuple) – the size of the padding. If is int, uses the same padding in all boundaries. If a 6-tuple, uses `(padding_left\text{padding\_left}padding_left, padding_right\text{padding\_right}padding_right, padding_top\text{padding\_top}padding_top, padding_bottom\text{padding\_bottom}padding_bottom, padding_front\text{padding\_front}padding_front, padding_back\text{padding\_back}padding_back)` Input:  $(N, C, D_{in}, H_{in}, W_{in})$  or  $(N, C, D_{in}, H_{in}, W_{in})$  or  $(C, D_{in}, H_{in}, W_{in})$ . Output:  $(N, C, D_{out}, H_{out}, W_{out})$  or  $(N, C, D_{out}, H_{out}, W_{out})$  or  $(C, D_{out}, H_{out}, W_{out})$ , where

## Function Signature

---

```
class torch.nn.ZeroPad3d(padding)[source]#
```

Parameters

Shape:

```
extra_repr()[source]#
```

Return type

**Returns:**

str

## Code Examples

---

```
>>> m = nn.ZeroPad3d(3)
>>> input = torch.randn(16, 3, 10, 20, 30)
>>> output = m(input)
>>> # using different paddings for different sides
>>> m = nn.ZeroPad3d((3, 3, 6, 6, 0, 1))
>>> output = m(input)
```

## Detailed Documentation

---

# Documentation

Pads the input tensor boundaries with zero.

For N-dimensional padding, use `torch.nn.functional.pad()`.

padding (int, tuple) – the size of the padding. If is int, uses the same padding in all boundaries. If a 6-tuple, uses  $(padding\_left, padding\_right, padding\_top, padding\_bottom, padding\_front, padding\_back)$

Input:  $(N, C, D_{in}, H_{in}, W_{in})$  or  $(N, C, D_{in}, H_{in}, W_{in})$  or  $(C, D_{in}, H_{in}, W_{in})$

Output:  $(N, C, D_{out}, H_{out}, W_{out})$  or  $(N, C, D_{out}, H_{out}, W_{out})$  or  $(C, D_{out}, H_{out}, W_{out})$ , where

$D_{out} = D_{in} + padding\_front + padding\_back$

$H_{out} = H_{in} + padding\_top + padding\_bottom$

$W_{out} = W_{in} + padding\_left + padding\_right$

Return the extra representation of the module.

---

